

—**General Instructions:**

- The submission due date of this assignment is **Monday(16th March 2026) 11:59 PM**
- Write a report (i.e. in a word file) that illustrates your main solution steps including the best fitness values and the average values, plotted over generations (with and without elitism)
- Zip your code and the report in a file entitled **[YourName_YourID_AssignmentNumber]**
- This assignment should be delivered and discussed INDIVIDUALLY

—**Requirements:**

In your previous assignment, you wrote a Python program that implements the simple GA to solve the OneMax problem. You used binary representation (genes either 0 or 1), randomly initialized population of size 20 chromosomes, roulette wheel selection, one-point crossover with $p_{\text{Cross}} = 0.6$, bit-flip mutation with $p_{\text{Mut}} = 0.05$ and elitism of size 2. You made 10 runs with different random seeds, each for 100 generations. For each run, you had a vector saving the highest fitness and the average fitness in the population at each generation.

Part I:

Building upon your previous assignment (the same implementation details as mentioned above), solve the following optimization problem:

Maximize $F(x_1, x_2) = 8 - (x_1 + 0.0317)^2 + (x_2)^2$, where $-2 \leq x_1, x_2 \leq 2$.

Increase the population size to 100 chromosomes and compare the performance of your GA under the following:

- 1- Standard decoding and gray decoding.
- 2- Different precisions (different number of bits to encode the variables).

Part II:

Extend your implementation of Part I to solve the same optimization problem under the following constraint: $x_1 + x_2 = 1$.

The fitness after penalty is computed as $F(x_1, x_2) - |x_1 + x_2 - 1|$.